

How can evolutionary biology inform the practice of investing for systems change?

TWIST case studies demonstrate how you can work with evolved psychologies to leverage systems change in practice.

Introduction

Investing for systems change is often described as the effort to shift the underlying relationships, incentives, norms, and power arrangements that produce outcomes across a whole system, rather than funding isolated projects one by one. Yet the field remains early, fragmented, and still in the process of clarifying which strategies make systems change more likely to succeed. That makes it worth asking not only what the practice of investing for systems change aims to do, but also what the nature of the human beings it is asking to do it is.

This is where evolutionary biology can help provide insight into the environments in which our brains and bodies evolved, and what we are really adapted to. Importantly, it suggests that systems-change strategies are more likely to succeed when they are built with evolved human psychologies in mind, rather than ignoring or working contrary to them. Much of human psychology was shaped in small-scale, uncertain social worlds in which immediate threats, visible rewards, local reputation, kin ties, and repeated interaction mattered greatly for survival and reproduction^{1, 2}. Systems change investment, by contrast, often asks people to act across long time horizons, cooperate beyond familiar groups, and invest in complex systems that do not yield immediate, visible results. The gap between those demands and the environments in which many human traits were shaped is what the evolutionary literature helps illuminate^{2, 3}.

Evolutionary biology and what it suggests about human beings

Evolutionary theory, together with work on contemporary hunter-gatherer societies and cultural evolution, suggests human beings are adapted for flexible life in small-scale groups, but not for frictionless cooperation in large, anonymous, highly abstract systems. Humans evolved in conditions where life was often uncertain, mortality was high, and where attending to immediate threats and opportunities was highly adaptive. Importantly, biological evolution is slow, meaning that our underlying biology is much the same as when we were hunter-gatherers in these environments. This helps explain why immediate rewards, visible cues, and short-term security still motivate humans to act, even in our modern world^{4, 5, 6}.

It is important not to forget that humans are unusually cooperative, but that cooperation is scaffolded. It depends on social norms, trust and repeated interaction, reputation, punishment, and institutions that widen trust beyond close ties⁷. Evidence from hunter-gatherer societies such as the Hadza in Tanzania shows cooperation can extend beyond kin, but it is strongly shaped by factors such as social structure and repeated interaction⁸. The evolutionary point, then, is not that large-scale cooperation is impossible, but that it must be actively built and supported.

What investing for systems change requires

Investing for systems change often asks for exactly the kinds of behaviours that do not come most easily. It asks people to tolerate uncertainty, delayed returns, and keep investing in change when progress is slow and diffuse. For example, systems change investors are asked to accept more risk with unknown, long-term returns. Systems change investing also requires cooperation across sectors, geographical regions, and cultures with people they may not naturally trust. The complexity of systems change asks people to think in terms of feedback loops, indirect effects, and interdependence rather than singular causes and simple solutions^{9, 10}. This means investing for systems change is not only a structural or financial challenge. It is also a psychological one.

It is important to note that this discussion does not begin with Systems Change theory as a starting point; rather, it is an examination of the practice itself, specifically within the limited scope of some of the TWIST case studies. By looking at how practitioners are already navigating these challenges, we can use evolutionary biology to better understand *why* certain methods work. If evolved human tendencies include present bias, parochial trust, status sensitivity, and difficulty reasoning about delayed, nonlinear systems, then systems change work must either fight those tendencies or redirect them. More will get done with the latter.

Friction no.1: Why would people act to benefit future generations if they can't see the reward now?

A first friction is between the slow speed of systems change and the human pull towards immediate reward. Long-term systems change requires people to accept uncertainty, slower cumulative progress, delayed returns, and trust-building. Yet humans are often more responsive to immediate benefits and threats than to distant, abstract outcomes. Van Vugt and Griskevicius argue that environmental and collective problems are especially hard because people prefer immediate rewards to delayed ones and often ignore problems they cannot directly see or feel⁶. Griskevicius et al. add that harshness and mortality cues can strengthen present-oriented strategies, particularly among those who grew up in more resource-scarce settings⁴. Resistance to long-term systems change is not just a failure of will, but a predictable response to uncertainty, distance, and weak immediate feedback.

Read through this lens, one barrier to systems change is that it often asks people to bear clear short-term costs in exchange for uncertain long-term gains. Patience, trust-building, and slow cumulative progress may be normatively desirable, but they do not map neatly onto the kinds of incentives humans find easiest to sustain.

Regen Melbourne and Hawai'i Investment Ready

TWIST case studies Regen Melbourne and Hawai'i Investment Ready (HIR) can be read as attempts to work with this evolved cognitive bias to actually reduce this friction. In [Regen Melbourne](#), a long-term initiative with the aim of remaking Melbourne into a resilient, regenerative city, large systemic goals are translated into more tangible forms - with 'Earthshot' themes (declarations of intent) and specific project aims. One example is 'making the Yarra swimmable by 2030', which breaks a vast systems challenge into a nearer-term narrative target. Adding to this, a shared framework such as Doughnut Economics (combining a social foundation with an ecological ceiling) gives participants an immediate sense of clarity and orientation. As something more tangible, Regen Melbourne's events provide immediate returns through connection, shared learning, and visible momentum, as people come together across the city to ask questions, share food, and work out what comes next. These are all examples of ways to reduce the psychological distance associated with long-term change.

[HIR](#) is an initiative to transform Hawai'i's economy to be resilient, regenerative and community-rooted, increasing social and environmental opportunities. It appears to do something similar to Regen Melbourne through trust and relational safety. Its slow, trust-first process, such as starting with months of relationship building, can be read as a way of making long-term cooperation feel socially secure in the present. Additionally, the Hawai'i Capital Scan gives participants immediate visibility into where capital is flowing or blocked; this instant strategic clarity works as a near-term reward, making collaboration feel advantageous right away rather than only in the distant future. Rather than asking actors to ignore immediate incentives, HIR changes what those incentives are. Status, belonging, mutual recognition, and shared values become part of the return people can see now, whilst deeper systemic gains are still unfolding. In evolutionary terms, it does not try to fight people's natural focus on immediate rewards, but tries to give that focus a better direction.

Friction no.2: Why would those within a system trust those from outside that system?

A second friction is between the wide cooperation systems change requires and the more bounded cooperation humans often find easiest. Systems change frequently depends on collaboration across cultures, sectors, ideologies, and institutional roles. It often asks people to share power with those they do not know well and whose interests they may not fully trust.

Evolutionary theory suggests that this is not a trivial demand. Human cooperation is real, but it is structured by kinship, reciprocity, reputation, and institutions^{1, 7}.

Work on parochialism is especially relevant here. Romano et al. find across 17 societies that people tend to trust and cooperate more with ingroup than outgroup members¹¹. Evidence from small-scale societies similarly suggests that cooperation is easier where repeated interaction and stable social ties make reciprocity viable¹². The implication for systems change is that the wider cooperation it requires should not be assumed to be easy. It must be built through conditions that make cross-group trust, visibility, and reciprocity more adaptive - to work with the friction, rather than against it.

Pioneers of Our Time and CO_

The TWIST case studies, *Pioneers of Our Time* and CO_ can be understood as practical attempts to work with this friction, and create conditions for wide-group cooperation where they are not naturally occurring. [Pioneers of Our Time](#) aims to regenerate the Muga Valley in Catalonia as a bioregional system, led by outsiders who are not originally from the region. These leaders act as bridge-builders by extending kinship to the local community and creating opportunities to work directly *with* the friction of collaborating beyond strangers. By using tools like social presencing theatre, an embodied method to surface cultural nuances; systems mapping, a visual framework for identifying leverage points; as well as shared celebration, such as local festivals and a community documentary, they build a shared water-centered identity that allows people living there to come together beyond their immediate kin. In this setting, long-term cooperation becomes less like a sacrifice for strangers and more like taking care of their own wider group.

[CO_](#), a systems-change studio, operates on a similar logic of leading as outsiders while ensuring participants see clear value in the process. By placing stakeholders into shared learning processes, it creates the repeated contact and reputational visibility necessary for trust. A key move is making contributions visible through systems mapping, which lowers the evolutionary fear of free-riding and raises the social value of participation. Whilst these strategies do not provide all the answers, they demonstrate how trust can be built, even by outsiders as mediators, by leveraging mechanisms humans already respond to - reciprocity and status. Ultimately, the goal is not to assume spontaneous selflessness, but to make cooperation safer and more rewarding by acknowledging and navigating the inherent friction of the system.

Why is it so hard for us to understand complexity?

Systems change requires people to think beyond singular systems, linear causes, and immediate effects, yet human cognition often struggles when outcomes are delayed, nonlinear, and generated by many interacting parts. Sterman demonstrates that people struggle to

understand dynamic systems because they focus on the problem directly in front of them, failing to track feedback loops and the downstream effects of their own actions. Because cause and effect are often separated in time and space, we rely on mental models that are too simple to capture systemic reality^{13, 14}. Johnson and Levin argue that modern crises are unlike the problems human psychology evolved to handle; our brains are better suited to immediate, visible threats than to delayed, interconnected ones¹⁵. This means the challenge is not just that people resist change - it is that they often struggle even to perceive the system they are being asked to change.

Recognising these cognitive hurdles, the TWIST case studies leverage specific frameworks to scaffold human understanding and make complexity manageable. For instance, doughnut economics provides a clear visual compass for a 'thriving' system, defining a safe and just space between social foundations and ecological ceilings, whilst systems mapping helps actors visualise the web of relationships and identify leverage points. Practitioners also use social presencing theatre as an embodied method to surface cultural nuances that intellectual models miss, alongside shared learning and months of relationship building to create the social security necessary for long-term cooperation. Furthermore, tactics like capital scans help simplify financial complexity by identifying available resources for the collective. It is important to recognise that these tools are inherently limited - a map must draw a boundary at a certain time point and cannot remain as fluid as the system itself - but they still serve as essential maps to help humans navigate a complex world.

Conclusion

An evolutionary perspective does not undermine the practice of investing for systems change, but sharpens it. It suggests that systems change in practice should take human psychology seriously as both a constraint and a design resource. If people are drawn toward immediate rewards, then long-term change needs nearer-term meaning, visibility, and social return. If people trust bounded groups more readily than strangers, then wider cooperation needs strong relational and institutional scaffolding. If people struggle to reason about delayed and interconnected systems, then shared sense-making, narrative frames, and visible feedback become essential.

The TWIST case studies evidence how practitioners can work with these perspectives to reduce the friction of change. Regen Melbourne and Hawaii Investment Ready address the pull toward immediate rewards by turning abstract, long-term goals into tangible narrative targets and nearer-term social returns like belonging and shared values. Pioneers of Our Time and CO_ demonstrate how to build trust between strangers by using tools like systems mapping to make everyone's contributions visible and by creating a shared identity that makes people feel like they are part of the same wider group. By providing opportunities for actors to acknowledge and work directly with these frictions - rather than treating it as an insurmountable barrier - these initiatives show that change happens most effectively when we redirect our natural habits rather than fighting them. Evolutionary biology, therefore, offers a helpful insight to support

practitioners in building initiatives that are more likely to succeed by navigating the reality of human nature.

Papers:

1. Henrich J. The Origins and Psychology of Human Cooperation. *Annu Rev Psychol.* 2020;71:335-362.
2. Van Vugt M, Li NPL. The Evolutionary Mismatch Hypothesis: Implications for Applied Psychology and Behavioral Economics. *Curr Opin Psychol.* 2020;32:32-37.
3. Li NPL, van Vugt M, Colarelli SM. The Evolutionary Mismatch Hypothesis: Implications for Psychological Science. *Curr Dir Psychol Sci.* 2018;27(1):38-44.
4. Griskevicius V, Tybur JM, Delton AW, Robertson TE. The influence of mortality and socioeconomic status on risk and delayed rewards: A life history theory approach. *J Pers Soc Psychol.* 2011;100(6):1015-1026.
5. Amir D, Jordan MR, Rand DG. An uncertainty management perspective on long-run impacts of adversity: The influence of childhood socioeconomic status on risk, time, and social preferences. *J Exp Soc Psychol.* 2018;79:217-230.
6. Van Vugt M, Griskevicius V. Naturally Green: Harnessing Stone Age Psychological Biases to Foster Environmental Behavior. *Soc Issues Policy Rev.* 2014;8(1):1-32.
7. Henrich J. Cultural group selection, coevolutionary processes and large-scale cooperation. *J Econ Behav Organ.* 2004;53(1):3-35.
8. Apicella CL, Marlowe FW, Fowler JH, Christakis NA. Social networks and cooperation in hunter-gatherers. *Nature.* 2012;481(7382):497-501.
9. Sterman JD. Learning from Evidence in a Complex World. *Am J Public Health.* 2006;96(3):505-514.
10. Korteling JE, van de Sande MCE, Schädler AMJ, Brouwer AM. Cognitive bias and how to improve sustainable decision making. *Front Psychol.* 2023;14:1100523.
11. Romano A, Balliet D, Yamagishi T, Liu JH. Parochial trust and cooperation across 17 societies. *Proc Natl Acad Sci U S A.* 2017;114(48):12702-12707.
12. Smith D, Dyble M, Thompson J, Major-Smith K, Page AE, Chaudhary N, et al. Camp stability predicts patterns of hunter–gatherer cooperation. *R Soc Open Sci.* 2016;3(7):160318.
13. Sterman JD. Modeling Managerial Behavior: Misperceptions of Feedback in a Dynamic Decision Making Experiment. *Manage Sci.* 1989;35(3):321-339.

14. Sterman JD. Learning from Evidence in a Complex World. *Am J Public Health*. 2006;96(3):505-514.
15. Johnson DDP, Levin SA. The tragedy of cognition: psychological biases and environmental inaction. *Curr Sci*. 2009;97(11):1593-1603.